University of Aldmere
School of Civil and Environmental Engineering

THESIS PROPOSAL · PhD qualifying review

Learning Where to Look

Adaptive Sparse Sensing for Street-Level Urban Heat Monitoring

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Summary

IN ONE SENTENCE

This thesis asks how a city can learn, from a few hundred cheap sensors, where to put the next one so that street-level heat is mapped well enough to protect the people most exposed to it.

Heat is the deadliest weather hazard in European cities, yet official temperature records come from a handful of airport and rooftop stations that miss the street canyons, courtyards and bus stops where people actually spend hot afternoons. Low-cost sensor networks close part of this gap, but they are placed by convenience rather than by need: on lampposts that happen to have power, in neighbourhoods that volunteer.

The proposed research treats sensor placement as a sequential learning problem. A Gaussian-process model of the urban temperature field, built on land-cover and morphology covariates, scores each candidate site by how much it would reduce uncertainty where vulnerable residents live. The thesis develops this exposure-weighted acquisition rule (RQ1), tests whether models trained in one city transfer to another (RQ2), and measures the gain in a live deployment of 60 mobile units in Aldmere (RQ3).

Preliminary work on 214 existing sensors shows that exposure-weighted placement reaches the mapping error of random placement with roughly 40 % fewer sensors.

KEYWORDS	urban heat, Gaussian processes, active learning, sensor networks, environmental justice
QUESTIONS	three research questions, Chapter 3
DURATION	four years, February 2025 to January 2029
FUNDING	Aldmere Doctoral Fellowship; field units from the City of Aldmere Climate Office
DATA	214 municipal sensors (2021–2025), 3 m land cover, LiDAR building heights, census vulnerability index

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1 Introduction

During the July 2022 heatwave, air temperature at Aldmere's official airport station peaked at 36.1 °C. On the same afternoon, volunteer sensors in the dense Old Harbour district recorded 39.4 °C, and a hospital two kilometres inland admitted three times its usual number of patients with heat stroke (Rautio et al. 2023). The official record was not wrong; it simply measured a different place. Urban heat varies over tens of metres, with street orientation, tree cover and building height, while the networks that inform public warnings resolve tens of kilometres (Oke et al. 2017).

Cheap connected sensors have made dense street-level networks affordable, and many cities now run them (Chapman et al. 2017; Meier et al. 2017). Yet where those sensors go is

rarely a design decision. They are mounted where a power outlet exists, where a school agreed to host one or where a citizen-science group happens to be active. The result is a network that is densest in affluent, well-organised districts and thinnest where older residents live alone in top-floor flats, the population most at risk (Hsu et al. 2021).

1.1 Problem statement

This thesis addresses the question of *where to put the next sensor*. Given an existing network, a model of the temperature field and a map of who is exposed, which site would most improve the information used to protect people? Posed this way, sensor placement becomes an instance of active learning: choosing the observation that is most valuable under a model, then updating the model and choosing again (Krause et al. 2008; Settles 2012).

Two features make the urban case harder than the textbook one. First, value is not uniform: an error of one degree over an empty car park matters less than the same error over a care home. Second, the model must work before much local data exists, because a city deploying its first network has nothing to learn from but other cities.

1.2 Scope

The thesis concerns near-surface air temperature during summer afternoons and nights in mid-latitude European cities. Surface temperature from satellites is used only as a covariate. Humidity and thermal comfort indices are outside the main line of work, although the field campaign records them for later use.

2 Background and related work

2.1 Mapping urban temperature

Urban climatology has long described the city as a mosaic of local climate zones whose thermal behaviour follows from built form and land cover (Stewart and Oke 2012). Statistical maps of air temperature interpolate between stations using these covariates, most often through land-use regression or kriging with external drift (Hjort et al. 2016). Recent work replaces linear trends with gradient-boosted trees or neural networks, gaining accuracy but losing the calibrated uncertainty that a placement rule needs (Venter et al. 2021).

Gaussian processes keep that uncertainty explicit. With a kernel that combines spatial distance and covariate similarity, they predict both a mean temperature and a variance at every unobserved location (Rasmussen and Williams 2006). Sparse approximations make them tractable for city-wide grids of a few hundred thousand cells (Hensman et al. 2013).

2.2 Optimal and adaptive sensor placement

Choosing sites to maximise information is a classical problem in spatial statistics. Mutual information between observed and unobserved locations is submodular, so a greedy rule achieves at least $(1 - 1/e)$ of the optimum (Krause et al. 2008). Adaptive variants place sensors one batch at a time and refit the model between batches, which matters when the kernel itself is uncertain (Garnett 2023).

These methods treat all locations as equally important. Cost-sensitive and task-driven variants exist in robotics and hydrology, where the objective is a downstream decision rather than a map (Marchant and Ramos 2014), but none has been applied to the distribution of heat exposure across a population.

2.3 The gap

Three things are missing from current practice. No placement rule weights uncertainty by exposure; no study tests whether a placement policy learned in one city transfers to another; and no adaptive scheme has been validated in a live urban deployment, where sensors fail and access is negotiated site by site. The research questions in the next chapter address each gap in turn.

3 Research questions

The gap described above leads to one overarching aim: *to make the placement of urban heat sensors an explicit, learnable decision that weights error by human exposure.* Three questions break this aim into parts that can each be answered within a thesis chapter.

> RQ1 **Exposure-weighted acquisition**
>
> *How should a sequential placement rule trade off reduction in predictive variance against the number of vulnerable residents near each candidate site, and how much does such a rule improve mapping error where it matters most compared with variance-only and random placement?*

> RQ2 **Transfer between cities**
>
> *Can a temperature model and its placement policy, learned from a dense network in one city, guide the first deployment in a city with almost no street-level data, and which covariates make that transfer succeed or fail?*

> RQ3 **Field validation**
>
> *In a live season-long deployment of mobile sensors that are moved every two weeks, does adaptive placement deliver the gains predicted in simulation once sensor drift, vandalism and access constraints are included?*

The questions build on each other: RQ1 supplies the rule, RQ2 asks how far it generalises, and RQ3 tests it against the untidiness of a real street. Each question has a falsifiable outcome, stated as a hypothesis in Chapter 4.

4 Methodology

4.1 Temperature model

Let $T(\mathbf{s}, t)$ be air temperature at site $\mathbf{s}$ and hour t. The model writes it as a covariate trend plus a spatially correlated residual,

$$T(\mathbf{s}, t) = \mathbf{x}(\mathbf{s})^\top \boldsymbol{\beta}_t + f(\mathbf{s}, t) + \varepsilon, \qquad f \sim \mathcal{GP}\big(0, k_\theta\big), \tag{4.1}$$

where $\mathbf{x}$ holds sky-view factor, impervious fraction, canopy cover and distance to water at 3 m resolution, and k_θ is a product of a Matérn kernel in space and a periodic kernel in hour of day. The model is fitted with stochastic variational inference on 512 inducing points.

4.2 Exposure-weighted acquisition (RQ1)

For a candidate site $\mathbf{c}$, the acquisition score is the reduction in exposure-weighted predictive variance across the grid $\mathcal{G}$,

$$a(\mathbf{c}) = \sum_{\mathbf{g} \in \mathcal{G}} w(\mathbf{g}) \big[\sigma^2(\mathbf{g}) - \sigma^2(\mathbf{g} \mid \mathbf{c})\big], \tag{4.2}$$

with weights w from the census vulnerability index (residents over 75, living alone, top-floor flats). Batches are chosen greedily, which keeps the near-optimality guarantee of the unweighted case.

Hypothesis H1. At equal sensor counts, rule (4.2) lowers exposure-weighted RMSE by at least 20 % relative to variance-only placement, with no more than 5 % loss in unweighted RMSE.

4.3 Transfer between cities (RQ2)

The model trained on Aldmere is applied to two partner cities with dense reference networks, Lyre and Westhaven, after withholding all their data. Placement is simulated from zero to 150 sensors, comparing a cold start, a transferred prior and a prior adapted by importance weighting on covariate distributions (Sugiyama et al. 2007).

Hypothesis H2. A transferred prior halves the number of sensors needed to reach 1.0 K weighted RMSE in the target city.

4.4 Field validation (RQ3)

Sixty battery-powered units with radiation shields are deployed from June to September 2027. Every two weeks, the twenty with the lowest marginal value are moved to the sites the rule proposes, subject to a list of permitted mounting points agreed with the city.

Hypothesis H3. Field gains are at least two thirds of the gains predicted by simulation for the same period.

5 Preliminary results

A first version of the model was fitted to the 214 municipal sensors operating in Aldmere between 2021 and 2025. Leave-one-out validation on summer afternoons gives an RMSE of 0.82 K and a mean 90 % interval coverage of 0.88, close to nominal. Figure 5.1 compares placement rules in a retrospective simulation: starting from 20 random sensors, each rule adds sensors from the remaining pool and the error is measured on all withheld sensors, weighted by exposure.

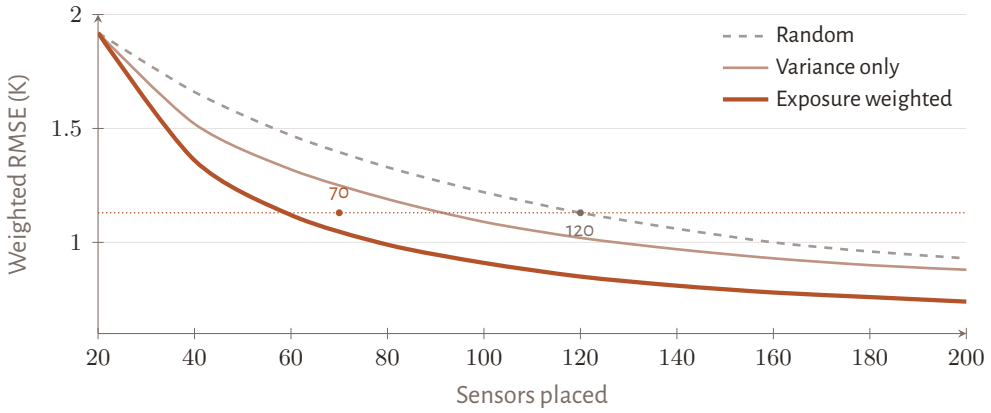


Figure 5.1. Retrospective placement in Aldmere, mean of 50 random starts. The dotted line marks the error random placement reaches with 120 sensors; the weighted rule reaches it with about 70.

Table 5.1 summarises the same experiment at 100 sensors. The weighted rule gives up little unweighted accuracy, which supports the trade-off assumed in H1, though the simulation draws only from sites that already host a sensor and so understates the choice available in a real deployment.

Table 5.1. Placement rules at 100 sensors (Aldmere, summer afternoons).

Rule	Weighted RMSE	Unweighted RMSE	Coverage
Random	1.22 K	0.97 K	0.86
Variance only	1.09 K	0.89 K	0.88
Exposure weighted	0.91 K	0.92 K	0.89

6 Thesis chapter plan

The thesis follows a monograph structure with three core chapters, one per research question, each written to stand as a journal article. Table 6.1 lists the chapters with their status at the time of this review.

Table 6.1. Planned chapters, their questions and their status.

No.	Chapter and content	Status
1	**Introduction.** Heat exposure, sensing gaps, aims and thesis outline.	outline
2	**Background.** Urban climate mapping, Gaussian processes, adaptive design.	draft
3	**Data and model.** Aldmere network, covariates, the model of Eq. (4.1) and its validation.	draft
4	**Exposure-weighted placement.** The rule of Eq. (4.2), theory and simulations; answers RQ1.	in progress
5	**Learning across cities.** Transfer to Lyre and Westhaven; answers RQ2.	planned
6	**A season on the street.** Field campaign and its lessons; answers RQ3.	planned
7	**Conclusions.** Synthesis, guidance for city climate offices, limitations and future work.	planned

Chapter 4 is planned as a submission to *Environmental Modelling & Software* in spring 2027, Chapter 5 to *Urban Climate*, and Chapter 6 to *Nature Cities* or a comparable venue, depending on the strength of the field result.

7 Timeline

The programme runs over eight semesters. Figure 7.1 shows the work packages, the milestones and the point of this review. The field campaign is fixed to summer 2027 because it needs a full warm season; the other tasks have a semester of slack before submission.

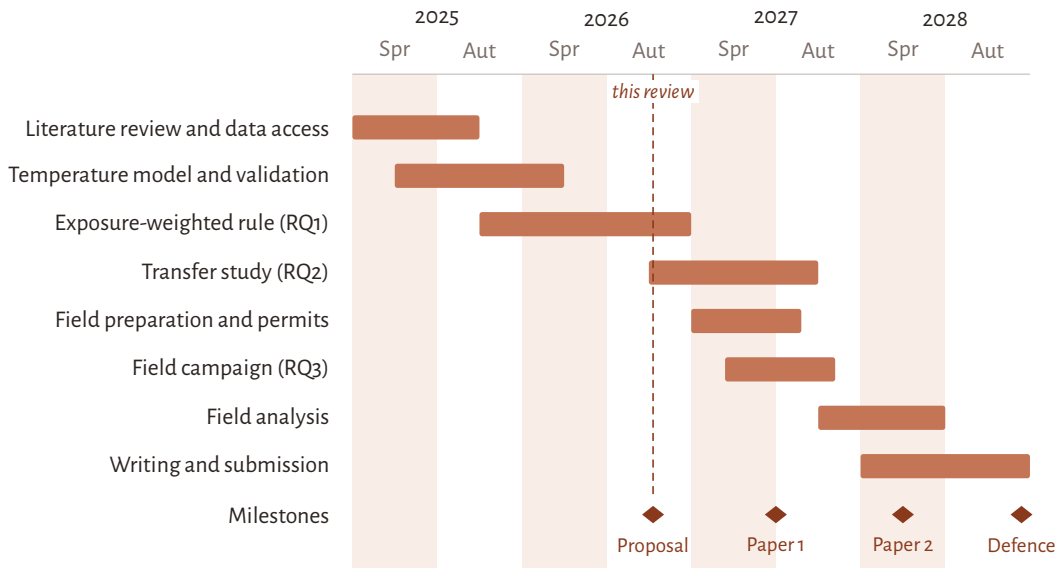


Figure 7.1. Work plan by semester. Shaded columns are spring semesters; diamonds mark milestones.

8 Risks and mitigation

Table 8.1 lists the risks judged most likely to delay the thesis. Likelihood and impact are rated low, medium or high; the mitigation column states what is already in place.

Table 8.1. Main risks to the programme.

Risk	Likel.	Impact	Mitigation
Mounting permits are delayed beyond May 2027	med	high	List of 140 pre-approved lampposts agreed with the city in 2026; campaign can start with these alone.
A cool summer in 2027 gives few heat days	med	med	Nights above 20 °C are frequent even in cool summers; the analysis also uses the 2028 municipal data.
Partner city withdraws data access	low	high	Data-sharing agreements signed with both cities; Westhaven data are already archived.
Sensor loss or vandalism above 20 %	med	low	Ten spare units; tamper-resistant mounts; loss is itself recorded as a result for RQ3.
Weighted rule shows no gain over variance-only	low	med	A null result is publishable and still answers RQ1; Chapter 5 does not depend on it.

9 Expected contributions

If the plan succeeds, the thesis will contribute the following.

C1 **A method.** An exposure-weighted acquisition rule for spatial sensing, with its near-optimality guarantee and an open implementation that a city climate office can run on a laptop.

C2 **Evidence on transfer.** The first systematic test of whether urban temperature models and placement policies carry over between cities, with guidance on which covariates to trust.

C3 **A field result.** A measured, not simulated, estimate of what adaptive placement gains on real streets over a full summer, including the losses that simulations ignore.

C4 **An open dataset.** Four months of street-level temperature and humidity from 60 relocating sensors, released with metadata under an open licence.

Beyond these, the work gives cities a concrete answer to a question they already face each spring: with a fixed budget, where should the next sensor go so that the next heat warning reaches the people who need it?

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